

EXAM QUESTION PAPER

College/ Institute	Engineering, Design and Physical Sciences		
Department	Computer Science		
Exam Author(s)	Dr. Derek Groen		
Module Code	CS3005		
Module Title	Digital Media and Games		
Month	May	Year	2022-23
Exam Type (Full/Resit)	Full	Format	WISEflow
Duration	3 hours		
Number of questions	19		
Question Instructions	Answer ALL questions in WISEflow. [Write other instructions here] When you have finished your exam, please click on the "Go to submission" (top right of your screen) to submit your answers; this will not be done for you if you forget or run out of time. Go to submission		
Calculators permitted?	Yes		
Can students include drawings/ diagrams?	Yes, in the last Section. This can be done by using the web camera on laptops, or using Google Slides.		
Permitted reference materials (including external websites)	A selected Google Slides working document is made available for each student.		
Required stationery	Pen and paper (only paper will be provided).		

By continuing beyond this point, you confirm that you have read the information and instructions above and understand the conditions of this examination.

1. In terms of designing interactions and interfaces, which is meant by a "cohesive visual style"?

A	a visual style that combines the latest elements from the interface research literature.
B	a visual style that contains a large amount of documentation, and provides access to an extensive library of other reference material.
C	a visual style that unifies similar operations, provides consistent interaction patterns and simplifies by limiting the use of very unusual elements.
D	a visual style that takes into account the location and language skills of the player, and adapts to travel between geographic locations.

2. Within the game Sim City, players are able to design and build customised cities. When players choose to customize the looks of individual buildings, which of Scott Osterwell's Four Freedoms are being expressed?

A	Freedom to Explore
B	Freedom of Identity Play
C	Freedom to Fail
D	Freedom to Effort

3. In the game Diablo players defeat monsters alone or together. It is also possible for the players to group together in guilds, and for guilds to rise in the guild ranking. What kind of player interaction pattern does the guild mechanic in Diablo have?

A	Single Player
B	Team Competition
C	Cooperative Play
D	Multilateral Competition

4. Boundaries as a formal element has a very specific role within a computer game. How would you describe this role?

A	The Boundaries formal element defines which game areas players can access, and which they cannot.
B	The Boundaries formal element defines the extent to which the game controls can affect the movement and actions of the player character.
C	The Boundaries formal element defines what is considered ethical within the game context and what is not.
D	The Boundaries formal element separates everything that is part of the game from everything that is not.

5. In the context of Prototype Game Design, developer Shannon McWhistle from Shaman Games has developed a physical prototype for the battle sequence in a game called "Mysterious Quest". Mysterious Quest intended to be a 3D game on a regular PC, with some turn-based strategy/battle elements.

Shannon McWhistle went to the DIY store to get large quantities of cardboard and cheap paint, and subsequently spent five weeks putting together a virtual castle setting to try and see whether his battle turn sequence works. Overall, he managed to make the prototype spending just a hundred pounds of materials, and the twelve playtesters he invited to play the prototype are having a good day trying out his turn sequence.

At the end of the day, Shannon looks at the feedback from the players, and realizes that although the playtesters enjoyed the setting, the turn sequence doesn't work as he had hoped and will need to be improved.

Now the question is as follows: while Shannon was making and using the prototype to get feedback about his turn sequence, he violated an important rule of thumb about prototyping. Which one?

A	He did not minimize time to first failure
B	His prototype did not have a clearly stated purpose.
C	His prototype was uninspiring.
D	His prototype was not falsifiable

6. In the context of AI, what is the main drawback if the AI of a Non-Player Character is too predictable?

A	The NPC becomes easy to beat.
B	The NPC becomes difficult to understand.
C	The NPC becomes overly powerful
D	The NPC can be easily "gamed" by the player.

7. In the context of simulation in games, why is a second order integration scheme often more accurate than a first order integration scheme?

A	A second order integration scheme gives a larger reduction in error when the time step size is reduced.
B	A second order integration scheme incorporates more particles.
C	A second order integration scheme uses an adaptively refined mesh which increases its accuracy.
D	None of the above, in almost all cases a first order integration scheme will be more accurate than a second order integration scheme.

8. Monochrome images have a range of drawbacks compared to 32-bit color images. But what is an advantage of monochrome images?

A	Increased colour depth.
B	Less storage space required.
C	Support for animations.
D	Superior resolution.

9. In the context of 3D graphics pipeline, which tasks are typically performed before the 3D scene is projected to 2D?

A	model, scale, rotate, translate, lighting, shading, texturing, rasterization.
B	model, scale, rotate, translate, lighting, shading, texturing, clipping.
C	model, scale, rotate, translate, lighting, shading, texturing.
D	There is no projection to 2D in the a typical 3D graphics pipeline

10. In the context of player engagement, a game offering players to buy a silver key when they are in front of a silver chest and ran out of silver keys is an example of:

A	Loss aversion.
B	Endowed Progress.
C	Triggering.
D	Cheating.

11. In the context of designing game interactions and interfaces, please describe the main difference between a diegetic interface element and a spatial interface element.

B

I

U

0 / 10000 Word Limit

12. On the topic of Game AI, for Finite State Machines, what does a Transitional Function describe?

B

I

U

0 / 10000 Word Limit

13. Games with laggy or choppy gameplay are not enjoyable, so performance optimisation is highly useful in game development. Name three particularly effective ways to optimise game performance, such that games run smoothly.

B

I

U

0 / 10000 Word Limit

14. In 3D graphics, different lighting and shading techniques can be used to calculate how light shines on a 3D object. The following image is a rounded donut shape.



What type of lighting and shading is used to visualise this object? And how does this type of shading work, resulting in this kind of look?

B

I

U

0 / 10000 Word Limit

15. Meaning, Mastery and Autonomy are key elements that drive player engagement. In the case of mastery one important aspect is that the player fails from time to time. Why is this important?

B

I

U

0 / 10000 Word Limit

You have been hired by The Unbridge Cultural Centre (TUCC) to build an educational game for 12-14 year olds (early secondary school) to teach them about different cultures in the world, and raise awareness of the diversity of people that live in Unbridge today, in terms of culture, language, religion, traditions and lifestyle. At this early stage you are required to produce a **report** detailing your initial ideas.

TUCC would like the game to address two key requirements:

- Requirement 1: The players should develop knowledge about a wide range of cultures in the world today (e.g., the origin of these cultures, key traditions and characteristics in terms of e.g. arts, language, activities and/or cuisine, and main influences that have shaped each culture). They should understand a range of subject-specific vocabulary relating to different cultures, major movements and civilisations, and gain some understanding of historical geography.
- Requirement 2: The game should lead to players gaining more interest in diverse cultures, and have the ability to recognise and acknowledge persons and items with different cultural backgrounds.

The organisation would like you to sketch out a game concept that consists of several components that are consistent with each other. These include a game idea and concept (question 16), a recommendation in terms of choice of platform and controller (question 17), an overview of core issues in the concept (question 18), and a story board + architectural diagram of your game concept. (question 19).

16. Please present your game idea and concept here, in response to the requirements and constraints given by the company.

B

I

U

0 / 10000 Word Limit

17. Please provide recommendations in terms of choice of platform and controller here.

B

I

U

0 / 10000 Word Limit

18. Please provide an overview of core issues in your game concept, citing references to the literature where relevant. Given the closed environment in which this exam takes place, incomplete references are acceptable, as long as they are sufficiently descriptive for readers to locate the content.

B

I

U

0 / 10000 Word Limit

19. Please upload a story board as well as and an architectural diagram for your game.

For this assignment you will be able to use Google Slides, which you can access using the "External resources" button on the upper right side of the screen. The google slide will open in a new tab. You can then create the prototype document, download this to your laptop and reupload it here in WISEflow, all within the parameters of the locked down environment. Alternatively, you are free to use any offline tool you have at your disposal, or to draw the diagrams on paper and upload them using your laptop camera.

The upload tool here supports PowerPoint, OpenOffice, PNG, JPG, GIF or PDF formats.

Change File

No file chosen

Supported file formats: PDF, JPG, GIF, PNG, Powerpoint, Open Office

0 / 12 File Limit